SE Experiment-5

(Batch-A/A3)

Team Details

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Aim:

To draw a **UML Activity Diagram for an Inventory and Stock Management System (ISMS)**, representing the flow of inventory, sales, purchasing, stock monitoring, user management, and external service interactions.

Theory:

An Activity Diagram is a UML behavioral diagram used to represent the flow of activities and control within a system. It shows the sequence of actions, decisions, alternative paths, parallel activities, and termination of a process. Activity diagrams are useful for understanding the dynamic behavior of a system and for representing complex business workflows in a simple visual form.

A swimlane activity diagram further divides the workflow into different lanes based on the responsibility of each actor or system component. This makes it clear which participant is responsible for performing a particular activity. In the HRMS activity diagram, separate swimlanes are used for the Guest, HRMS, Payment Gateway, OTA/Channel Manager, AI Concierge Service, Notification Service, Smart Lock Controller, Receptionist, Hotel Manager, Finance Staff, and System Administrator.

The diagram uses standard UML elements such as an initial node to indicate the beginning of the workflow, activity nodes to represent operations, decision nodes to represent conditions, control-flow arrows to show the sequence of activities, fork and join nodes to represent parallel operations, and a final node to indicate the completion of a workflow.

Problem Description:

An Activity Diagram is a UML behavioral diagram used to represent the flow of activities and control within a system. It describes the sequence of operations, decision points, alternative flows, and termination of a process. It is useful for modeling the dynamic behavior of a system and understanding complex workflows in a structured manner.

A swimlane activity diagram divides the workflow into different lanes according to the responsibility of each actor or system component. In the Inventory and Stock Management System (ISMS), separate swimlanes are used for the Administrator, Manager, Sales Personnel, Warehouse Staff, ISMS (System), and External Services. This makes it clear which participant performs each operation and how they interact with the system.

The diagram uses standard UML elements such as an initial node to represent the start of the workflow, activity nodes for system operations, decision nodes for conditional branches, control-flow arrows to represent the sequence of activities, and a final node to indicate completion.

The workflow includes activities such as user authentication, role identification, sales order processing, inventory operations, purchase order management, stock-level monitoring, analytics updates, and low-stock alert generation. The system also interacts with external services for payment processing, marketplace order synchronization, notifications, voice processing, and demand forecasting.

Implementation:
The activity diagram is implemented using **UML swimlanes** to represent the interaction between the different users, the ISMS, and external services.

The workflow begins with the **ISMS validating login credentials**. If the credentials are invalid, an error message is displayed and the user is directed back to the login validation process. If the credentials are valid, the system identifies the user's role and displays a **role-based dashboard**.

The **Administrator** performs user and role management, supplier management, purchase order approval, and system configuration. The **Manager** views the dashboard, generates forecast reports, reviews reorder suggestions, and approves or rejects purchase orders.

The **Sales Personnel** create sales orders and scan or select products before checking stock availability. If stock is unavailable, a stock-out message is generated. If stock is available, the system processes the payment, generates an invoice, and updates the sales order.

The **Warehouse Staff** select an inventory operation and can choose between the standard UI and voice-based interaction. For voice interaction, the spoken command is processed through the Speech-to-Text Service, interpreted by the system, and confirmed before the corresponding stock record is updated.

After the relevant operations, the ISMS **synchronizes information across channels, checks reorder levels, and updates analytics data**. If the stock level falls below the reorder point, a **low-stock alert** is generated. The system also communicates with external services for online order synchronization, payment processing, notifications, voice processing, and demand forecasting, completing the overall inventory management workflow.

Conclusion:
The UML Activity Diagram provides a clear representation of the overall workflow of the **Inventory and Stock Management System**. The use of swimlanes effectively identifies the responsibilities of administrators, managers, sales personnel, warehouse staff, the ISMS, and external services. Decision nodes represent alternative conditions such as stock availability and reorder levels, while the different system and external-service interactions demonstrate how inventory operations are integrated with payment, notification, voice-processing, marketplace, and forecasting services. Thus, the activity diagram provides a structured view of the dynamic behavior of the ISMS and helps in understanding the flow of inventory and stock-related operations.

Activity Diagram:

